

## PANEL MOUNTED WEIGHING INDICATOR AND HIGH SPEED TRANSMITTER



Incorporating the latest up-to-date electronics, the Model XT2000 is a high performance display indicator and high speed data transmitter suitable for static and dynamic weighing. It is specially designed for process weighing systems, as well as all types of machinery with weighing as part of their function. Because of its panel mount design and versatility of digital input/output and data transmission options, it is ideally suited to connection to PLCs, PCs and remote systems. Using the optional flexible cover, IP65 protection is possible.

Typical applications include: weighing and level measurement of tanks and silos, industrial scales, filling and dosing machines, animal weighing, high speed sorting scales and multi-head weighers. The integrated check weigher application, coupled with the high speed data capture rate of 2400 measurements/second, also makes the XT2000 ideal for use in check weighing and other dynamic applications.

First start-up and commissioning can be done easily using PC software. Calibration is possible either with real weights or by using the load cell sensitivity in mV/V (theoretical calibration).

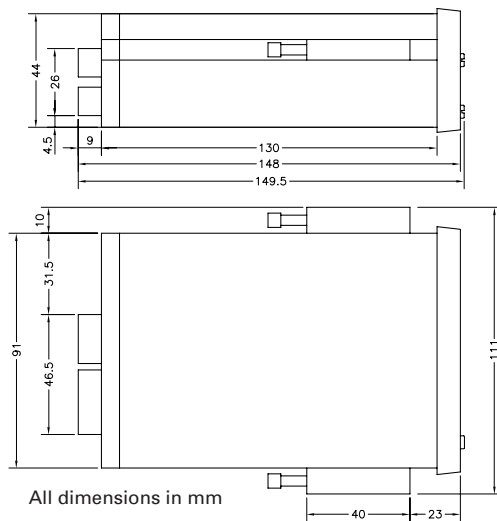
The XT2000 is approved according to OIML R76 and EN 45501 as a Non-Automatic Weighing Instrument.

Five versions are available; the basic version has digital I/O, RS232 and RS485; 4 further versions add either analog outputs, PROFIBUS, PROFINET or Ethernet/IP communications.

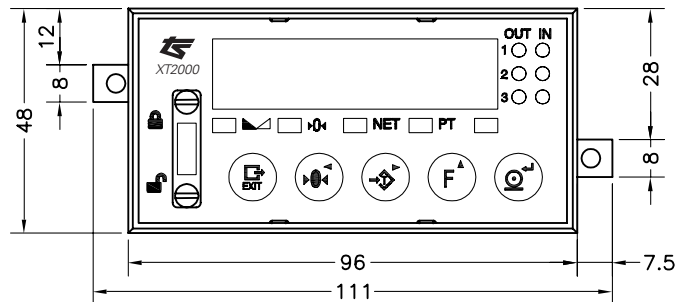
- Bright, 6 digit, 10mm LED display
- High resolution with 24-bit ADC and 16,000,000 internal divisions
- 2400 measurements per second
- Configuration and calibration using software or integral keypad
- Calibration protection by software or mechanically
- Powers up to 8 X 350Ω or 16 X 700Ω load cells
- RS232 and RS485 (Modbus RTU and ASCII) and optional PROFIBUS, PROFINET or Ethernet/IP
- Configurable analog outputs: 0-5V, 0-10V, 0-20mA or 4-20mA, with galvanic isolation
- 3 digital opto-isolated inputs and 3 digital outputs to relay, with status LEDs
- Panel cut-out size 92mm W x 45.5mm H
- 10-28 V DC power supply. Optional AC-to-DC adapter available (DIN rail mounted)

# XT2000

technical specification...



All dimensions in mm



## XT2000 Indicator / Transmitter

| Load Cell Connection         |                                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| Load cell sensitivity range  | ± 3.9 mV/V                                                                                             |
| Resolution                   | ADC 24-bit; 16 000 000 internal divisions;<br>100 000 external divisions<br>Display -99 999 to 999 999 |
| Measuring rate               | Internal: 2400 readings per second<br>Transmission: 600 readings per second                            |
| Linearity error              | ≤ 0.01% of measuring range                                                                             |
| Thermal stability            | Zero: 150 nV / °C max.<br>Span: 3.5 ppm / °C max.                                                      |
| Load cell excitation voltage | 5.0 V DC                                                                                               |
| Minimum load cell resistance | 43Ω (8 load cells of 350Ω or 16 load cells of 700Ω)                                                    |

| Inputs & Outputs                                             |                                                                |
|--------------------------------------------------------------|----------------------------------------------------------------|
| Serial communications                                        | RS-232 bi-directional                                          |
|                                                              | RS-485 half-duplex                                             |
|                                                              | Protocol: Own protocol, Modbus (RTU & ASCII)                   |
|                                                              | Transmission rates:<br>4800 to 115200 bauds selectable         |
| Logic inputs                                                 | 3 opto-isolated inputs with status LED                         |
| Logic outputs                                                | 3 relay outputs max. 30V AC – 30V DC,<br>100mA with status LED |
| Analogue Output<br>(XT2000-SO-AN version)                    | Output with galvanic isolation, DAC of 16 bits                 |
|                                                              | Output voltage: 0-10V                                          |
|                                                              | Output current: 0/4 - 20mA (<500Ω)                             |
| PROFIBUS communication<br>(XT2000-SO-PB version)             | PROFIBUS DP-V0 (Slave)                                         |
| PROFINET communication<br>(XT2000-SO-PN version)             | PROFINET I/O Real-time (Class A)                               |
| Ethernet/IP communication<br>(XT2000-SO-ETHERNET-IP version) | Ethernet/IP                                                    |

| Additional Data             |                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power supply                | 10V to 28V DC                                                                                                                                                                                         |
| Power consumption           | 4W maximum (6W for PROFIBUS, PROFINET and Ethernet/IP versions)                                                                                                                                       |
| Operating temperature range | -20 °C to 50 °C                                                                                                                                                                                       |
| Storage temperature range   | -30 °C to 60 °C                                                                                                                                                                                       |
| Nominal shipping weight     | 0.25kg                                                                                                                                                                                                |
| Mounting                    | Panel mounting, cut-out size<br>92mm W x 45.5mm H                                                                                                                                                     |
| IP rating                   | IP40. IP65 using optional flexible<br>cover (XT2000-IP65-COVER)                                                                                                                                       |
| Operator interface          | Keyboard: 5 membrane keys                                                                                                                                                                             |
|                             | Display: 6 digit LED, 10mm high                                                                                                                                                                       |
| Options                     | With serial outputs only<br>(model XT2000-SO)                                                                                                                                                         |
|                             | With serial and analog outputs<br>(model XT2000-SO-AN)                                                                                                                                                |
|                             | With serial outputs and PROFIBUS<br>(XT2000-SO-PB)                                                                                                                                                    |
|                             | With serial outputs and PROFINET<br>(XT2000-SO-PN)                                                                                                                                                    |
|                             | With serial outputs and Ethernet/IP<br>(XT2000-SO-ETHERNET-IP)                                                                                                                                        |
| Approvals                   | OIML R76                                                                                                                                                                                              |
|                             | EN 45501:2015 (NAWI)                                                                                                                                                                                  |
| Optional Power Supply       | DIN rail mount power supply                                                                                                                                                                           |
|                             | Universal voltage 100-240V AC                                                                                                                                                                         |
|                             | 50/60Hz. 24V DC output                                                                                                                                                                                |
|                             | Maximum Power 15W, capable<br>of driving 3 off XT2000 (-SO<br>and -SO-AN version) or 2 off<br>XT2000 (-SO-PB, -SO-PN and<br>-SO-ETHERNET-IP versions).<br>Part number XT-PS.<br>Shipping weight 0.1kg |



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## THAMES SIDE SENSORS LTD

Unit 10, io Trade Centre, Deacon Way,  
Reading, Berkshire RG30 6AZ, UK

tel: +44 (0) 118 941 1387

sales@thames-side.co.uk

www.thames-side.com

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Our policy is one of continuous product enhancement. We therefore reserve the right to incorporate technical modifications without prior notification.



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